

SangWook Woo

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EDUCATION

B.S. in Electronic and Information Engineering, Soongsil University

- Period: Mar/2020 - Aug/2026 (Expected)
- GPA: 4.28 / 4.5 (144 credits completed)
- Major GPA: 4.25 / 4.5 (111 major credits completed)
- Two-year leave of absence due to mandatory military service

Relevant Coursework

- Embedded / System: Embedded Systems, Microprocessor Applications, Digital FPGA Design, Operating Systems, Data Communication, Computer Architecture
- Circuit / Hardware: Digital System Design, Circuit Analysis, Electronic Circuits, Applied Electronic Circuits, Digital Signal Processing, Signals and Systems, Semiconductor Devices, Electromagnetics, Transmission and Propagation

AWARDS

- Hyeongnam Science Award - Bronze Prize, Soongsil University College of Engineering, Dec/2025
- [AIX2025] Deep Learning Hardware Design Competition - Encouragement Award, Seoul National University Next-Generation Semiconductor Research Group, Jun/2025
- 2021 Soongsil Capstone Design Competition - Encouragement Award, Soongsil University, Sep/2021

SCHOLARSHIP

- National Science & Engineering Scholarship, Mar/2024 - Sep/2025
- Heunghan Foundation Scholarship, Mar/2021 - Sep/2021
- Academic Excellence Scholarship (Quarter), Sep/2020

EXPERIENCE

RISC-V Core-based AXI-Lite/APB/AXI-Stream Mini MCU RTL Integration

- Period: Apr/2026 - Present
- Designed a mini MCU RTL system integrating an RV32I 5-stage pipeline core, AXI-Lite interconnect, APB peripherals, AXI-Stream DMA, PLIC-lite interrupt controller, and UART bootloader
- Implemented hazard, forwarding, branch, CSR, and trap paths; converted the CPU DBus into an AXI-Lite master for I-SRAM, D-SRAM, DMA control, and APB peripheral access
- Built a RISC-V GCC-based firmware flow including C firmware build, linker script, startup code, and UART loader packet generation to run RAM applications without regenerating the FPGA bitstream
- Verified UART RX -> DMA S2MM -> D-SRAM -> DMA MM2S -> SPI TX data movement and PLIC interrupt handling; confirmed 64x64 RGB image end-to-end transfer with byte mismatch 0 and pixel mismatch 0

SystemVerilog UVM-based RTL Verification Practice

- Period: May/2026 - Present
- Built UVM 1.2 testbench structures for Adder, RAM, FIFO, UART, SPI, and I2C example RTL blocks
- Practiced sequence/item, driver, monitor, scoreboard, and coverage components; applied reference-model-based readback checks and basic SVA/assertions
- Separated Basic practice flows based on VCS/Verdi from Communication practice flows based on Vivado/XSim, organizing sources, logs, and reports by simulator flow
- Focused on transaction-based stimulus/check separation, scoreboard pass/fail criteria, and coverage-oriented verification rather than claiming production-level verification ownership

Undergraduate Research Intern supervised by Prof. Incheol Song

- Period: Dec/2025 - Jan/2026
- Designed and implemented an AXI-to-APB bridge using Verilog-HDL
- Implemented burst-to-single transfer handling, address decoding, and APB slave selection
- Performed simulation and functional verification with Verilog testbenches across read/write scenarios

FPGA Video Filtering Accelerator Project

- Period: Sep/2025 - Nov/2025
- Built a real-time 3x3 video filtering system on Ultra96-V2 (Zynq) in an HW/SW co-design flow, streaming OV5640 camera input directly to a TFT-LCD
- Designed Window3x3 and LCD/output-buffer control logic in Verilog, applying double buffering, CDC synchronization, and frame-drop-based sync control
- Implemented PS-side C code for camera initialization and filter control; verified integrated system performance at 81.97 FPS

SoC Chip Power and Size Estimation Training

- Period: Aug/2025
- Studied basic methodologies for estimating SoC chip size and power at an early design stage by considering process, memory configuration, area, and power factors

PCB Circuit Design & PSpice Designer Training

- Period: Jul/2025 (60 hours)
- Completed PCB Circuit Design / Capture / PSpice Designer training at 9ES Campus (Cadence Authorized Training)
- Practiced schematic design, simulation, and analysis using Capture and PSpice Designer
- Gained hands-on experience in PCB layout, circuit implementation, and simulation-based design verification

Microprocessor Applications

- Period: Mar/2025 - Jun/2025
- Studied ARM processor architecture
- Served as team leader for an image processing project
- Optimized image processing performance with ARM Assembly

Operating System Project

- Period: Mar/2025 - Jun/2025
- Solved system-level tasks in the xv6 operating system
- Studied core operating system concepts including virtualization, concurrency, and persistence

System Programming Project

- Period: Mar/2025 - Jun/2025
- Developed a SIC/XE assembler with C and Java
- Developed a SIC/XE simulator in Java to support instruction execution and memory management

AGV Path Planning and Deadlock Resolution Simulation

- Period: Mar/2025 - Nov/2025
- Implemented a control algorithm to reduce collision and deadlock issues among multiple AGVs in an unmanned parking environment
- Combined WHCA*, WFG/SCC-based deadlock detection, and Partial CBS-based replanning logic
- Compared performance with A* and D* Lite on realistic and extended parking maps, validating throughput and computational efficiency in dense environments
- This project led to the Hyeongnam Science Award - Bronze Prize

[AIX2025] Deep Learning Hardware Design Competition

- Period: Feb/2025 - May/2025
- Studied quantization algorithms, CNNs, FPGA resource utilization, Vivado, and AXI protocols
- Designed and verified a custom CNN accelerator in Verilog, optimizing for limited FPGA resources and high-throughput performance
- Analyzed related research papers to better understand AI accelerator design and implementation flows

Design Compiler-based Digital Logic Synthesis Training

- Period: Feb/2025
- Practiced basic Synopsys Design Compiler concepts, command usage, constraint application, and synthesis-result review flow

Embedded System Project

- Period: Sep/2024 - Jan/2025
- Gained hands-on experience with ATtiny and STM32 microcontrollers using Microchip Studio and STM32CubeIDE
- Controlled sensors, switches, LEDs, motors, and touch panels using C, HAL libraries, and direct register access
- Analyzed datasheets and designed circuits under limited pin constraints
- Implemented interrupts, timers, SPI, and I2C-based control

Undergraduate Research Intern supervised by Prof. Incheol Song

- Period: Dec/2024 - Jan/2025
- Designed and implemented the AES algorithm using Verilog-HDL
- Developed RTL-level hardware design and integrated an APB interface for register-level communication
- Performed simulation and functional verification using Verilog testbenches

Semiconductor and Display Engineering Class Project

- Period: Nov/2024 - Dec/2024
- Studied semiconductor processes for one semester
- Target circuit: LDO (Low-dropout regulator)
- Designed the circuit layout to minimize area while reflecting design rule constraints
- Planned an optimized fabrication process

Small-sized Elevator Control System Project

- Period: Sep/2024 - Dec/2024
- Served as team leader
- Designed and implemented a miniature elevator system using the STM32F103
- Programmed motor control, button input, floor display, and floor queue management logic in C

Digital System Design Class Project

- Period: Oct/2024 - Nov/2024
- Served as team leader
- Developed FIR filter architecture using Verilog-HDL at the RTL level
- Coordinated task division, integration, and simulation using testbenches
- Received an A+ in the course

Full-Custom IC Design Course

- Period: Jul/2024 - Oct/2024 (60 hours)
- Completed the Cadence Full-Custom IC Design & Term Project course at Soongsil University

- Learned Cadence Virtuoso, Schematic, Layout, Spectre, and Assura
- Designed basic logic gates and a 4-bit full adder/subtractor using full-custom layout and schematic
- Performed DRC/LVS checks and circuit simulation with Spectre

Soongsil Capstone Design Competition

- Period: Jun/2021 - Sep/2021
- Developed a COVID-19 prevention automatic door system
- Utilized Arduino and Raspberry Pi for sensor control and automation
- Implemented mask detection, thermal temperature screening, and CCTV monitoring using cameras and infrared sensors

Soongsil Electronic Exhibition (SEE) Club

- Period: Mar/2021 - Present
- Participated in the annual SEE exhibition project
- Instructed new SEE members in C programming
- Provided academic and embedded systems project mentoring to junior members

CERTIFICATION

- OPIc IH, Mar/2025 - Mar/2027
- Computer Specialist in Spreadsheet & Database Level-1, Jul/2023 - Present
- Korean Driver's License (Class 2), Dec/2022 - Present
- Craftsman Information Processing, May/2019 - Present

SKILLS

Language

- Verilog-HDL | Intermediate: AXI4-to-APB Bridge, APB-AES, CNN Accelerator, FIR Filter, FPGA Video Filtering, RV32I-based SoC RTL. AMBA bus interfaces, datapath/control logic, testbench-based functional verification.
- SystemVerilog | Intermediate: interface, package, struct/enum, assertion syntax. RISC-V SoC integration and UVM transaction-level testbench practice.
- UVM | Basic: sequence/item, driver, monitor, scoreboard, and coverage structure. Example benches for Adder/RAM/FIFO and UART/SPI/I2C.
- C | Intermediate: STM32/AVR MCU control, RISC-V firmware, xv6 coursework, embedded projects. Sensor, motor, LCD, interrupt, timer, SPI/I2C control, and PS-side driver code.
- ARM Assembly | Intermediate: ARM architecture-based image processing optimization, C-to-assembly conversion, and performance comparison.
- Python | Basic: automation scripts, data processing, verification helpers, and experiment support scripts.
- Java | Basic: SIC/XE assembler and simulator implementation, basic object-oriented structure.

Tool

- Vivado | Intermediate: RTL synthesis/implementation, simulation result analysis, timing/resource review, AXI-based IP integration, Tcl and batch-script automation exploration.
- STM32CubeIDE / Microchip Studio | Intermediate: STM32/AVR MCU control, debugging, peripheral control, embedded development environment setup.
- Linux Environment | Intermediate: basic commands, file management, project build environment setup.
- Git | Intermediate: personal/team project version control, branch-based task division, history management.
- Cadence Xcelium / ModelSim | Basic: RTL simulation, waveform analysis, Verilog/SystemVerilog testbench checking.
- Synopsys VCS / Verdi | Introductory: compile/run flow, simulation log review, and waveform/debug flow in UVM basic practice.
- Vitis | Basic: PS-side driver code and AXI4-Lite control implementation in the FPGA Video Filtering project.
- Visual Studio | Basic: C/C++ assignment and project development environment.
- Keil | Basic: ARM Assembly image processing code writing, debugging, and performance comparison.
- Cadence Virtuoso / Capture / PSpice / PCB Editor | Basic: schematic, layout, circuit simulation, PCB layout, DRC/LVS basic verification flow.
- Synopsys Design Compiler | Introductory: synthesis basics, command usage, constraint application, and test flow.
- MS Office (Word, Excel) | Intermediate: document writing and data organization.

Others

- Hands-on experience with oscilloscopes and pulse generators
- Hands-on experience with PCB soldering and board-level measurement/validation